FACULTÉ DES SCIENCES DE LA SANTÉ
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FACULTY OF HEALTH SCIENCES
SCHOOL OF REHABILITATION SCIENCES

August 23, 2026

Subject: Submission to *PCI Neuroscience*

Dear Recommender,

We would like to briefly provide some background on our preprint "*Habitual physical activity and motor cortical neurophysiology: a systematic review with meta-analyses of transcranial magnetic stimulation measures*" (https://doi.org/10.51224/SportRxiv.815), as its previous peer-review process is relevant to the current submission.

The manuscript was previously submitted to a nonprofit, society-published scientific journal, underwent peer review, and was rejected after the first round. We considered the three reviewers' comments useful and addressable. To avoid wasting the effort invested by the reviewers, an increasingly pressing issue in our publishing system, we decided to address their comments. To make this progress transparent, we have made our point-by-point response to their comments on GitHub (https://github.com/Boisgontier-Lab/TMS-MetaAnalysis).

For two of the reviewers, we have provided our responses without reproducing their comments, out of respect for confidentiality and because our responses provide sufficient context to understand the issues raised. The third reviewer signed his report and gave us permission to share his comments. It turned out that this reviewer (Dr. Nicholas P. Holmes, University of Birmingham, UK) also serves as a Brain Stimulation Editor for *Cortex* and was highly positive about the manuscript. Among his comments, he wrote: "The systematic review and meta-analysis methods and results are excellent, very thorough, using the best-available tools and procedure. This is better than I would do myself […]".

After receiving the email informing us of the rejection, he invited us to submit our work to *Cortex*, writing that they would be "very happy to entertain this work" after we addressed the reviewers' comments and suggesting that we mention his invitation in the cover letter. We were very grateful for this invitation. However, *Cortex* is published by Elsevier, and for several years we have made a strong effort not to contribute our work to for-profit journals whose revenues are not substantially reinvested in the academic system. We therefore respectfully declined the invitation and chose instead to submit the manuscript to *PCI Neuroscience*.

We hope the manuscript will be considered on the basis of the question it addresses, the rigor of the methods, and the knowledge it contributes to the field, rather than the statistical significance of its results. We believe that null results are valuable to the scientific literature, particularly when they arise from a rigorous synthesis of

the literature and are supported by equivalence testing. Such results can help prevent researchers from continuing to invest substantial time and resources in questions for which meaningful effects may be unlikely, while encouraging more theoretically grounded questions or alternative methodological approaches. In the present case, the results are informative because they challenge the assumption that transient neurophysiological effects of acute exercise necessarily translate into stable differences associated with habitual physical activity.

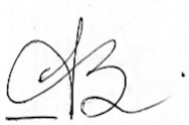
Further, the associations synthesized in our meta-analysis were not reported in the original publications. The included studies had collected measures of habitual physical activity and TMS-derived motor neurophysiology, but none reported estimates of the associations between these variables. We therefore recovered previously unreported association estimates from statistical information in the published articles, open datasets, and data obtained directly from authors, and synthesized them across studies. In this sense, our study represents a secondary quantitative analysis of existing data that generated new empirical evidence on a question that had not previously been examined. Despite substantial interest in the effects of acute exercise on cortical neurophysiology, the relationship between habitual physical activity and resting motor cortical neurophysiology had remained essentially unexplored.

Our work was preregistered in Prospero (CRD420251016029), and the data and analysis scripts are publicly available on Zenodo (https://doi.org/10.5281/zenodo.22069859).

We hope that you will find that both the question addressed and the approach taken in this manuscript make it a valuable contribution to *PCI Neuroscience*.

Thank you again for taking the time to consider our submission.

Best regards,
Matthieu Boisgontier



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